

Report
Of
Graphics Design and Product Prototyping for a Consumer
Toy Brand

Submitted
To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE

By
Roll Number of Student: 2410030504
Name of Student: NITIN KUMAR CHOUDHARY

Section: 3CSE26

Batch:
2026-27

1. CANDIDATE’S DECLARATION

I, Nitin Kumar Choudhary, Roll No. 2410030504, Section 3CSE26, do hereby declare that the internship report titled “Graphics Design and Product Prototyping for a Consumer Toy Brand” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

I further declare that this report has been prepared in accordance with the confidentiality undertaking given to Intelligentia Labs Pvt. Ltd., and that no proprietary artwork, source file or unreleased product detail has been reproduced without the express permission of the organisation.

Signature

Student Name: Nitin Kumar Choudhary

Roll No.: 2410030504

Date: _____

2. ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I wish to record my thanks to Intelligientia Labs Pvt. Ltd. for the opportunity to undertake this internship with the InventoJoy brand, and to my reporting manager, whose review and guidance shaped every deliverable described in this report. The habit of physically verifying a design before defending it, which recurs throughout this document, was learned directly from that supervision. I also thank the product and manufacturing teams for their patience in explaining constraints that were not obvious to me.

I extend my thanks to the faculty of the School of Computer Science and Engineering, IILM University, and to my internal guide, for their academic supervision and for their guidance in the preparation of this report.

I express my gratitude to my parents for their blessings.

Signature

Student Name: Nitin Kumar Choudhary

Roll No.: 2410030504

3. INTERNSHIP COMPLETION CERTIFICATE

ATTACH HERE: The internship completion certificate issued by Intelligientia Labs Pvt. Ltd. on company letterhead and signed by the organisation. Per the prescribed format, this page carries no page number. The certificate must be issued by the company — request it from your reporting manager or from the Director, InventoJoy, stating your name, role and exact start and end dates.

Certificate details to be confirmed by the organisation:

- Name: Nitin Kumar Choudhary, Roll No. 2410030504
- Designation: Graphics and Prototyping Intern, InventoJoy
- Organisation: Intelligientia Labs Pvt. Ltd., EPIP Site-5, UPSIDA, Greater Noida – 201306
- Period: 26 March 2026 to [end date]
- Issued and signed by: [name and designation, on company letterhead]

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4. PROJECT DESCRIPTION

4.1 Introduction

An industrial internship is the point at which academic preparation meets the constraints of commercial production. In a university setting a design problem is bounded and evaluated against a rubric. In an operating company the same problem is bounded by cost per unit, tooling limits, material availability, shipping dimensions, retail shelf conventions, statutory labelling and a delivery date that has already been promised to a customer. The transition between these two environments is the educational value of an internship, and it is the subject of this report.

This report documents an industrial internship undertaken as Graphics and Prototyping Intern at Intelligentia Labs Pvt. Ltd., which owns and operates the toy brand InventoJoy. The engagement was based at the company office at E-003, First Floor, EPIP Site-5, UPSIDA, Greater Noida – 201306, as a full-time work-from-office position commencing 26 March 2026 with a minimum contracted duration of three months.

The role sits between two disciplines that are usually taught separately but are inseparable in toy manufacturing. Graphic design governs how a product communicates with a child and with a purchasing parent. Physical prototyping governs whether the designed object can actually be produced, packed, shipped, opened and assembled. A toy that is beautifully illustrated but arrives in a crushed carton has failed; a carton that survives transit but cannot be opened by an eight-year-old has also failed. The internship was concerned with the space between those two failures.

The pairing of graphics and prototyping in a single job title is deliberate rather than accidental. A graphic applied to a flat artboard behaves differently once it is folded. Artwork crossing a crease will distort; a logo positioned too close to a cut line will be trimmed inconsistently across a production run; a colour that reads correctly on coated board will shift on an uncoated one. The only reliable way to find these problems before a print run is to build the object physically. Prototyping therefore functions as a verification step for graphic decisions, not merely as a stage of product development.

This report follows the format prescribed by the School of Computer Science and Engineering. Section 4.2 describes the host organisation, Section 4.3 states the problem the work addresses, Sections 4.4 and 4.5 set out objectives and scope, Section 4.6 documents the tools used, Section 4.7 presents the architecture of the design system, Section 4.8 presents the methodology together with a project executed end to end, and Section 4.9 records outcomes and learning.

4.2 Organization Profile

Intelligentia Labs Pvt. Ltd. is the corporate entity under which the internship was undertaken, operating from EPIP Site-5, an industrial area administered by the Uttar Pradesh State Industrial Development Authority in Greater Noida. The location places the design function close to the manufacturing and vendor ecosystem of the National Capital Region, which matters for a role that requires frequent physical review of samples, board stocks and printed proofs.

InventoJoy is the company's toy brand. The appointment letter describes it as a growing brand, and the responsibilities attached to the internship — packaging, dielines, inserts, stickers, prototypes, assembly guides and marketing creatives — are characteristic of a brand at the stage where product range and brand system are being built concurrently rather than sequentially.

This growth stage changes how a designer works. In a mature brand most design activity is application of an established system to a new product. In a growing brand it is partly application and partly system-building: templates, dieline libraries, colour standards and instructional iconography are created as a by-product of individual projects. Recognising this early was a useful adjustment, because a file built as a reusable template repays itself across the range while a file built as a one-off does not.

TO COMPLETE: Add a short factual paragraph on the company — year of incorporation, team size, product categories, target age bands and sales channels. Take these from the company website or ask your reporting manager; do not estimate.

4.2.1 Working environment and reporting structure

The engagement was a work-from-office position with working hours from 10:00 to 19:30, extendable according to operational requirements. Co-location matters for this role because prototyping is physical: board samples, printed proofs, folded mock-ups and assembled toys must be handled, folded, dropped and shown to colleagues. A dieline review happens around a table with the object present, not over a screen share.

Work was assigned and reviewed by the reporting manager, with periodic performance review as provided in the appointment letter. Three informal review mechanisms operated continuously and accounted for most of the feedback received: desk review of work in progress, physical review conducted around a mock-up or prototype, and cross-functional review by the product or manufacturing team.

4.2.2 Terms of engagement

Table 4.1: Summary of internship engagement terms

Parameter	Detail
Organisation and brand	Intelligentia Labs Pvt. Ltd. — InventoJoy
Designation	Graphics and Prototyping Intern

Parameter	Detail
Commencement	26 March 2026
Minimum duration	3 months, with a 7-day probation period
Working hours	10:00 to 19:30, subject to operational requirements
Mode and location	Work from office — EPIP Site-5, UPSIDA, Greater Noida – 201306
Review	Periodic performance review by the reporting manager
Notice period	30 days
Confidentiality	Non-Disclosure Agreement covering customer data, business strategies, source files, assets and operational processes
Restrictive covenant	Non-Compete Clause — 6 months post-departure, e-commerce and toy sector, within India
Progression	Pre-Placement Offer may be considered subject to performance

4.2.3 Confidentiality position adopted by this report

The Non-Disclosure Agreement expressly covers source files, assets and operational processes. Since these are the outputs of a graphics and prototyping role, the obligation determines what an academic report may legitimately contain. The position adopted is that method, reasoning, workflow, checklist and standard are described in full, because these constitute transferable professional knowledge; specific artwork, production dielines, vendor identities, costing and unreleased product detail are omitted, generalised, or reproduced only with the organisation’s written consent.

4.3 Problem Statement

The problem addressed by this internship is characteristic of a growing consumer manufacturing brand: design work cannot be judged on visual merit alone, because a single packaging design is simultaneously four different artefacts, each governed by a different constraint set.

It is a structural artefact, whose board grade, creases, glue flaps and closure determine whether it folds, holds its contents and survives a warehouse stack. It is a logistics variable, whose proportions fix cube utilisation, master carton fit and dimensional weight, and therefore a per-unit shipping cost on every unit ever sold. It is a regulatory instrument, carrying age grading, warning statements, manufacturer identification, country of origin, a conformity mark and a barcode, none of which are discretionary content. And it is a retail communication device, which has a few seconds on a shelf to establish hierarchy and signal an age band to a purchasing parent.

Optimising any one of these at the expense of the others is the commonest source of production failure. A graphic approved on screen may break when the board is folded. A carton that presents well on a shelf may be uneconomic to ship. An aesthetically

resolved layout may leave no legible space for a statutory warning. The consequence is discovered not at review but after tooling has been cut, at which point the cost of correction is a production cycle rather than an afternoon.

Table 4.2: Functions of toy packaging and their design implications

Function	Design implication
Containment	Internal volume, insert or tray design, immobilisation of parts to prevent rattle and abrasion in transit
Protection	Board grade and thickness, flute selection, cushioning, drop performance
Communication	Front-panel hierarchy, product imagery, feature callouts, brand system
Compliance	Age grading, warning statements, manufacturer details, country of origin, conformity marking, barcode
Retail performance	Shelf facing dimensions, stackability, hang-tab or shelf-ready format, planogram footprint
Logistics efficiency	Master carton fit, cube utilisation, pallet pattern, dimensional weight for e-commerce

A secondary problem, specific to the position of a junior designer, is sequencing. A designer new to a manufacturing environment tends to treat structural, regulatory and logistical requirements as review criteria applied to a finished creative proposal. Practitioners treat them as parameters admitted before creative work begins. The difference is not one of skill but of process, and it determines how much rework a project incurs.

4.4 Project Objectives

The objectives below derive from the responsibilities defined in the appointment letter and form the framework against which the work described in Section 4.8 was planned and reviewed.

1. To apply graphic design fundamentals to toy packaging and printed collateral — boxes, dielines, inserts and stickers — optimised for manufacturing, shipping and shelf display.
2. To develop working knowledge of structural packaging, specifically the construction and validation of dielines.
3. To assist in building physical prototypes and in testing products against functional and durability expectations.
4. To design instruction manuals and assembly guides that are intuitive, largely language-independent and appropriate to the comprehension level of a child user.
5. To collaborate with the product and manufacturing teams so that packaging and products submitted for production are Design-for-Manufacturing ready.

6. To support the marketing function with creatives suitable for digital and print deployment.
7. To develop professional competencies: version control of design files, structured vendor handover, responding to review feedback and working to production deadlines.

The five responsibility areas defined in the appointment letter map onto five work streams, which are used consistently throughout this report.

Table 4.3: Mapping of appointment-letter responsibilities to work streams

Responsibility as stated	Work stream	Principal competencies
Designing toys, packaging and graphics (boxes, dielines, inserts, stickers) optimised for manufacturing, shipping and shelf display	WS-1	Vector production, structural packaging, pre-press, colour management, shelf and logistics reasoning
Creating prototypes and helping test products	WS-2	Model making, 3D printing, materials handling, structured observation
Designing intuitive, child-friendly instruction manuals and assembly guides	WS-3	Technical illustration, pictogram systems, information sequencing
Collaborating with product and manufacturing teams for DFM-ready solutions	WS-4	Constraint analysis, technical communication, review discipline
Supporting marketing creatives for digital and print platforms	WS-5	Small-format composition, adaptation systems, asset management

4.5 Scope of the Project

This report covers the technical background necessary to understand the work, the organisational context in which it was carried out, the architecture of the design system operated, the methodology followed, one project documented end to end, and the outcomes and competencies arising. A single project is traced in full detail because method is demonstrated more convincingly by one artefact followed through than by a list of outputs.

The report does not cover commercial performance, supplier identities or commercial terms, or unreleased product concepts. Where a work item cannot be described specifically for confidentiality reasons, the class of problem and the method applied are described instead.

4.5.1 Limitations

- Confidentiality. The appointment letter imposes a Non-Disclosure Agreement covering source files, assets and operational processes — precisely the outputs of

this role. Proprietary artwork is therefore described, redacted or generalised rather than reproduced.

- **Measurement.** This was a design engagement, not a research study. No controlled experiment, statistically valid user test or instrumented measurement was conducted. Section 4.9 presents qualitative observations and a proposed evaluation framework rather than measured results, and every proposed metric is explicitly labelled as not measured.
- **Attribution.** Design work in a studio is collaborative and supervised. Deliverables described here were executed under the direction of the reporting manager and reviewed by the product and manufacturing teams. Design direction, vendor selection and final production approval sat with the organisation.
- **Duration.** A three-month minimum engagement limits exposure to the full lifecycle of a product from concept through to post-launch performance data.

4.6 Technologies and Tools Used

The tables below list the categories of tool relevant to a graphics and prototyping role and the purpose each serves. A confirmation column is provided so that the actual tools used can be recorded, since availability varies between organisations.

Table 4.4: Design and modelling software

Tool	Purpose in this role	Used
Adobe Illustrator / CorelDRAW	Vector artwork, dieline construction, logotype, pictograms, production-ready packaging files	[Yes / No]
Adobe Photoshop	Raster retouching, product photograph preparation, mock-up compositing	[Yes / No]
Adobe InDesign	Multi-page documents: instruction manuals, catalogues, trade sheets	[Yes / No]
Acrobat / pre-flight tools	Output verification, separation preview, overprint and bleed checking	[Yes / No]
Figma	Collaborative layout, digital creative iteration, review commenting	[Yes / No]
Fusion 360 / SolidWorks	Parametric modelling of components, tolerance and fit studies, export for 3D printing	[Yes / No]
Blender	Concept modelling and product visualisation for creatives	[Yes / No]
Slicing software	Preparation of models for additive manufacture	[Yes / No]

Table 4.5: Prototyping hardware and materials

Resource	Application	Used
FDM 3D printer	Functional and fit prototypes of components	[Yes / No]
Resin (SLA) printer	High-detail appearance models and fine features	[Yes / No]
Cutting plotter	Accurate cutting and creasing of board mock-ups from dielines	[Yes / No]

Resource	Application	Used
Board and substrate samples	Physical mock-ups at correct grammage and finish	[Yes / No]
Hand tools — scalpel, steel rule, bone folder, adhesives	Manual model making and mock-up assembly	[Yes / No]
Vernier caliper, steel scale, weighing scale	Dimensional and mass verification of prototypes and packs	[Yes / No]
Camera and lighting setup	Documentation of prototypes and photography for creatives	[Yes / No]

4.6.1 Technical foundations underlying the toolset

A dieline is a flat, dimensioned vector template defining how a sheet of board is cut, creased and assembled into a three-dimensional package. It is the interface between designer and die-maker, and the most error-prone artefact in packaging production, because an error in it surfaces only after tooling has been cut. Each line type is conventionally assigned a distinct spot colour so it can be separated and suppressed at output; misassigning a line type is a frequent cause of production failure.

Table 4.6: Dieline line types and common carton constructions

Element	Purpose or characteristic	Convention or application
Cut line	Where the board is cut through	Solid, dedicated spot colour
Crease line	Where the board is scored to fold	Dashed, separate spot colour
Perforation	Tear-open features and opening strips	Dash-dot, separate spot colour
Bleed	Artwork extended beyond the cut line	Typically 3 mm on every cut edge
Safety zone	Inner margin for critical content	Typically 3–5 mm inside cut and crease lines
Reverse tuck end	Tucks fold from opposite directions; economical die	Lightweight boxed items and small kits
Auto-bottom	Pre-glued base erects automatically; stronger base	Heavier contents, higher-volume lines
Window box	Die-cut aperture, usually with a clear film patch	Where product visibility drives purchase
Corrugated mailer	Self-locking single-wall construction for transit	Direct-to-consumer e-commerce shipping

[INSERT IMAGE HERE] — Export a generic dieline (not a proprietary one) and annotate the line types in Illustrator or PowerPoint.

Figure 4.1: Annotated dieline showing cut, crease, bleed and safety zones

Artwork is authored on screen in an additive RGB space but reproduced on press in a subtractive CMYK space, supplemented by spot colours. The conversion is not lossless: saturated blues, greens and oranges commonly fall outside CMYK gamut and shift. The practice adopted was to work in the destination colour space from the outset for brand-

critical elements and to specify brand colours as spot references with documented CMYK equivalents. Raster artwork for offset printing is prepared at approximately 300 pixels per inch at final size, with effective resolution checked at placed size rather than at nominal file dimensions.

Substrate and finish selection is a cost and performance decision. Folding boxboard grades differ in fibre composition, reverse-face whiteness and stiffness; corrugated flute profiles trade print quality against crush resistance. Lamination, coating, foiling and embossing each add cost and lead time, and each interacts with glue areas, which must be kept free of varnish and lamination.

4.6.2 Regulatory context

Toy safety in India is governed by a quality-control regime requiring conformity to prescribed Indian Standards and the corresponding conformity marking, covering mechanical and physical properties, flammability and migration of certain elements. Products intended for export are additionally assessed against destination-market standards. For a graphics role the regime manifests mainly through labelling: age grading, warning statements including small-parts warnings where applicable, manufacturer or importer identification, country of origin and the conformity mark. These are not discretionary content, and the correct practice is to place them on the dieline before creative layout begins rather than fitting them into whatever space remains.

VERIFY BEFORE SUBMISSION: Standards and quality-control orders are revised periodically. Confirm the exact standard designations, editions and marking requirements applicable to InventoJoy's products with your reporting manager or the company's compliance contact and cite them in the Bibliography. Do not state standard numbers you have not verified — an incorrect citation is worse than a general statement.

4.7 System Architecture

The design function operated as a layered system in which each layer constrains the one above it. Describing the work this way is not a retrospective abstraction: the layering determines the order of decisions, and reversing it is the principal cause of rework.

4.7.1 Layered architecture of the design system

Table 4.7: Layers of the design system and the constraints each imposes

Layer	Contents	Constraint imposed on the layer above
Requirement layer	Brief, product dimensions, audience, channel, deadline, approval path	Defines what counts as success; fixes the content inventory
Compliance layer	Age grading, warnings, manufacturer details, origin, conformity mark, barcode	Occupies defined, non-negotiable space on the artefact

Layer	Contents	Constraint imposed on the layer above
Structural layer	Dieline, carton construction, insert geometry, bleed, safety zones, glue areas	Fixes the canvas: panel sizes, fold positions, unusable margins
Graphic layer	Illustration, typography, colour, hierarchy, brand system	Must resolve within the space the three layers below leave
Verification layer	Mock-ups, prototypes, DFM review, pre-press checklist	Tests every layer against physical and process reality
Release layer	Version control, approval record, vendor handover pack, archive	Guarantees that what reaches production is what was approved

[INSERT IMAGE HERE] — Draw the six layers as a stack with arrows showing that each lower layer constrains the one above, and the verification layer feeding back into all of them.

Figure 4.2: Layered architecture of the design system

4.7.2 Artefact architecture

Each design file was constructed to a fixed internal architecture so that it could be handed to a vendor without explanation. The dieline occupies a separate, clearly named layer set to suppress at output. Mandatory compliance content occupies a defined region placed before creative layout. Artwork occupies the remaining layers, organised by panel rather than by element type, so that a change to one face of the carton does not require hunting through the file. Linked assets are packaged with the parent document, and fonts are embedded or outlined according to the vendor's requirement.

4.7.3 Version control and release architecture

All design files followed the pattern `PRODUCTCODE_ARTEFACT_VARIANT_vNN_STATUS.ext`, where the status token distinguishes WIP, REVIEW, APPROVED and PRODUCTION states. Only files carrying the PRODUCTION token were released to a vendor. This is a deliberate structural safeguard: a production error in a design studio is more often a version-control failure than a design failure, and a naming convention that makes an unapproved file visibly unapproved prevents that class of error without depending on anyone's memory.

- Version numbers increment on every substantive change and are never overwritten.
- Superseded versions are archived rather than deleted, so a reverted decision can be recovered.
- The approved version identifier is recorded in writing at the point of approval.

- The vendor handover pack comprises the production file with links packaged, the dieline on a separate suppressible layer, a flattened reference PDF, and a specification sheet stating substrate, finishes, colours and quantity.

4.7.4 Review gates

Each stage of the workflow exits through a gate with an explicit criterion. A gate is a constraint check rather than a consultation: nothing proceeds until each criterion is either met or a deviation is formally accepted by the responsible team.

Table 4.8: Review gates and exit criteria

Gate	Reviewer	Exit criterion
Brief confirmation	Reporting manager	All intake parameters resolved in writing
Concept selection	Reporting manager	One direction selected against brief criteria
Structural sign-off	Manager; manufacturing	Dieline physically folded and dimensions verified against the product
Design review	Reporting manager	Hierarchy, brand consistency and mandatory content approved
DFM review	Product; manufacturing	Every constraint satisfied or formally accepted as a deviation
Pre-press check	Designer; vendor pre-flight	All checklist items cleared
Final approval	Reporting manager	Approved version identifier recorded in writing

4.8 Methodology

The methodology is presented first as a general workflow and then as executed on a single project, so that the abstract sequence and its practical consequences can be read together.

4.8.1 The nine-stage workflow

1. Brief intake: objective, product, audience, channel, constraints, deadline and approval path.
2. Research and reference: category conventions, existing brand assets, competitor shelf presence, applicable compliance requirements.
3. Concept development: divergent options generated at low fidelity and evaluated against the brief rather than by preference.
4. Structural definition: dieline selected or constructed, with bleed, safety zones, glue areas and closure mechanism resolved.
5. Artwork development: graphics applied within the fixed template, mandatory content placed first.

6. Physical mock-up and iteration: print, cut, fold, assess; iterate on proportion, legibility and structural behaviour.
7. DFM and cross-functional review: constraint check with the product and manufacturing teams.
8. Revision and pre-press: feedback incorporated, pre-press checklist executed.
9. Handover and archive: vendor handover pack prepared, approval recorded, version archived.

Two features define the method. First, the dieline is fixed and physically folded before artwork begins; a dieline that has never been folded is an untested assumption, however carefully it was constructed on screen. Second, verification fidelity escalates only as far as the current question requires — a printed paper fold before a plotter-cut mock-up, a mock-up before a substrate comp, a comp before a vendor sample. A vendor sample is authoritative but slow and expensive; a folded sheet of paper costs fifteen minutes and, in practice, caught most layout and proportion errors.

[INSERT IMAGE HERE] — Draw the nine stages as a flowchart with two feedback loops marked: mock-up iteration returning to artwork development, and cross-functional review returning to structural definition.

Figure 4.3: End-to-end design and prototyping workflow

4.8.2 Brief intake

An incompletely specified brief is the most expensive failure mode in a design cycle, because it is discovered only after production effort has been spent. The intake practice was to confirm a defined parameter set in writing before starting: the objective and its success criterion; the product, variant and whether dimensions are final; the audience age band and the purchase decision-maker; the channel; whether the dieline is fixed, adapted or new; the full content inventory including copy, claims, statutory text, barcode and certification marks; production constraints of substrate, colours, finishes, vendor and quantity; and the deadline with its approval path.

4.8.3 Project executed: IJ-SC-01, a STEM construction kit

READING NOTE: This project is documented as a representative case of the work undertaken, using an anonymised product code, so that method can be traced through a single artefact. Before submission, align the product name, dimensions, board grades and outcomes with your own project and replace the figure placeholders with your own images.

The project covered the complete printed system for a STEM construction kit, referred to here as product code IJ-SC-01: a 42-part snap-together building set for the six-plus age band, sold both through retail shelf and through e-commerce marketplaces. The deliverables were a primary carton with dieline, an internal insert tray, a sticker sheet, a pictorial assembly guide, the master shipping carton specification, and a set of listing creatives.

Two constraints shaped every subsequent decision. First, the product sells through two channels with opposing requirements: retail shelf rewards a large, visually assertive facing, while e-commerce penalises volume through dimensional weight. Second, a 42-part kit with small components triggers small-parts labelling obligations and requires an insert that immobilises parts so they neither rattle nor abrade in transit.

Table 4.9: Project requirement summary for IJ-SC-01

Parameter	Requirement
Product	IJ-SC-01 — snap-together STEM construction kit, 42 parts
Age band	6 years and above
Channels	Retail shelf and e-commerce marketplace
Primary pack	Printed folding carton with product visibility
Internal	Insert or tray immobilising all parts; sticker sheet
Documentation	Pictorial assembly guide, minimal text dependence
Compliance	Age grading, small-parts warning, manufacturer details, country of origin, conformity mark, barcode
Transit	Master carton fit; survive e-commerce single-parcel handling
Constraint	Minimise pack volume without weakening shelf facing

4.8.4 Structural design and carton selection

Structural definition preceded artwork. Three constructions were evaluated against the requirement set.

Table 4.10: Carton construction options evaluated

Option	Advantages	Disadvantages	Assessment
Reverse tuck end	Lowest die and material cost; simple to erect by hand	Weaker base under load; opposing tuck directions untidy on the front face	Rejected — base strength inadequate for a parts-heavy kit
Auto-bottom with window	Strong pre-glued base; fast to erect; window gives product visibility on shelf	Higher die cost; window patch adds a process step and complicates recycling	Selected
Rigid lid and base	Premium presentation; reusable as storage	Substantially higher unit cost and greater pack volume	Rejected — cost and dimensional weight

The auto-bottom construction with a die-cut window was selected. The pre-glued base carries the load of a parts-heavy kit without the base failure characteristic of a tuck end, and it erects on the packing line without manual base folding. The window was sized to reveal the assembled hero model rather than the loose parts, which addresses the shelf requirement without enlarging the pack.

Internal dimensions were set from the insert tray envelope plus a tolerance allowance, and board thickness was accounted for at every fold rather than only at the outer dimensions — an omission that produces a carton which measures correctly flat but

binds when erected. Bleed and safety zones were applied to the vendor's specification, and cut, crease and glue areas were assigned to their conventional layers and spot colours.

[INSERT IMAGE HERE] — *Insert your flat dieline showing cut, crease, glue and window aperture on separate layers.*

Figure 4.4: IJ-SC-01 primary carton — flat dieline with line types annotated

4.8.5 Insert design and panel layout

A 42-part kit presents a containment problem that the outer carton cannot solve. Loose parts migrate in transit, abrade printed surfaces, and produce a rattle that reads at the point of sale as breakage. A die-cut folding board insert was selected over a vacuum-formed plastic tray and a moulded pulp tray: it is produced on the same substrate and by the same vendor as the carton, removing a supplier and a lead time from the critical path; it is compatible with the brand's direction on plastic reduction; and it can be revised at the cost of a die change rather than a mould. Its lower part-location precision was mitigated by grouping parts into compartments by size rather than locating each part individually. A lift tab allows a child to remove the tray without inverting the carton, and the sticker sheet sits flat beneath it where it is protected from creasing.

Artwork was developed panel by panel on the fixed dieline rather than across the flat sheet, because a composition that balances flat can be incoherent once folded. The front-panel hierarchy was resolved first, since it carries the commercial burden: the assembled hero model and product name lead, the brand mark follows, the age grading and part count sit third. The window aperture was treated as an element of the composition rather than as a hole cut into a finished layout; positioning it low and to one side allowed illustration and typography to occupy the optically dominant upper region while the physical product occupied the lower.

Two disciplines were applied without exception: no critical element within the safety zone of any cut or crease line, and mandatory content treated as fixed with creative content arranged around it. Statutory content was placed on the dieline before creative layout began. When mandatory content is placed last it occupies residual space, is set at minimum size on a busy ground, and becomes the weakest element on the pack; the rear panel was therefore laid out on a plain ground with the statutory block given a defined, uninterrupted region.

[INSERT IMAGE HERE] — *Insert the artwork-on-dieline flat and photographs of the folded carton, front and three-quarter.*

Figure 4.5: IJ-SC-01 — applied artwork on the dieline and assembled carton views

4.8.6 Prototyping and iteration

Verification followed the escalation principle described in Section 4.8.1. Each iteration was built to answer a specific question, and fidelity rose only when the question required it.

Table 4.11: Prototype iteration record for IJ-SC-01

Iteration	Method	What it revealed
v01	Desktop print, hand cut and folded	Front-panel typography fell within the crease safety zone; hierarchy competing between product name and hero model
v02	Desktop print, hand cut and folded	Revised window position improved the facing; insert tray envelope found to exceed the internal dimension
v03	Plotter-cut on representative board	Internal dimension corrected for board thickness; lift tab added to the tray
v04	Plotter-cut with parts loaded	Compartment depths adjusted; two compartments merged to reduce insert complexity
v05	Comp on intended substrate and finish	Brand colour shift observed on conversion; spot reference specified with a documented CMYK equivalent
v06	Vendor pre-production sample	Consistency of the production process — outcome to be recorded

TO COMPLETE: Replace the outcomes above with what your own iterations actually revealed, and record the outcome of the vendor sample. Iteration records are the most persuasive evidence in a design report because they show reasoning rather than results — be specific about what went wrong at each stage.

[INSERT IMAGE HERE] — Photograph successive mock-ups together, oldest to newest, in a single frame so the iteration is visible at a glance.

Figure 4.6: Prototype iteration sequence, first mock-up to final

4.8.7 Assembly guide design and validation

An assembly guide for a children’s product is a technical communication problem with a demanding audience: a child with limited reading ability, a parent under time pressure, or a user who does not read the printed language. The design response is to minimise dependence on text.

- Language-independent pictograms carry the instruction; text is supplementary rather than load-bearing.
- One action per step. Combining two actions into a single illustrated step is the commonest cause of assembly error.
- A consistent viewpoint across steps, since changing the viewing angle forces the reader to re-orient the object mentally.
- A parts inventory at the outset, each part illustrated at consistent scale with an identifier used throughout.

- Safety and adult-supervision notices placed before the step they relate to, not only on a cover.

For a 42-part kit the structure adopted was a parts inventory spread; a connection-principle page demonstrating how parts join, taught once rather than repeated; a step sequence for the hero model at one action per step and a fixed isometric viewpoint; and an alternative-builds section at lower instructional density.

Validation was carried out by handing the guide and the parts to a colleague unfamiliar with the product and observing silently while they assembled it. This method matters because a guide is written by someone who already knows how the product goes together, and the author is therefore almost incapable of detecting an ambiguous step. Every point at which the assembler hesitated was treated as a defect in the guide, not in the assembler. This is, in substance, a usability test conducted on a physical artefact.

TO COMPLETE: Record what your own validation revealed — which steps caused hesitation, and what you changed. This is the single most valuable paragraph in the section if you carried out the exercise.

[INSERT IMAGE HERE] — Insert the parts inventory page and a spread of consecutive assembly steps showing consistent viewpoint and one action per step.

Figure 4.7: Assembly guide — parts inventory and step sequence

4.8.8 Design for manufacturing review

The design passed through a joint review with the product and manufacturing teams before pre-press, conducted as a gate rather than a consultation.

Table 4.12: DFM constraints raised and design responses

Constraint raised	Origin	Design response
Glue flap dimension marginal for the specified board grade	Manufacturing	Flap width increased; dieline re-nested to absorb the change without additional sheet area
Window patch adds a process step and lead time	Manufacturing	Retained as an accepted deviation on shelf-visibility grounds; aperture reduced to limit patch material
Insert compartment count raised packing labour	Product	Two compartments merged; parts grouped by size rather than located individually
Heavy ink coverage encroaching on glue areas	Manufacturing	Artwork pulled back; glue areas kept free of varnish and lamination
Pack depth driving dimensional weight on e-commerce	Product	Depth reduced by revising insert compartment depths after iteration v04
Master carton fit leaving avoidable void space	Manufacturing	Primary pack footprint adjusted so units nest without void

The most consequential learning here concerned sequencing. A constraint discovered at review is a revision; the same constraint anticipated during design is simply a

parameter. Over the project the practice shifted from designing and then submitting for a constraint check, to consulting manufacturing at the structural definition stage.

4.8.9 Pre-press verification and handover

Pre-press was executed as a fixed checklist rather than as review by inspection, because the errors it catches are precisely those inspection misses.

Table 4.13: Pre-press verification checklist

#	Check	Verified
1	Document dimensions match the approved dieline exactly	[]
2	Bleed applied on every cut edge; no critical content inside a safety zone	[]
3	Dieline on a separate named layer set to suppress at output	[]
4	Cut, crease and perforation lines on their correct spot colours	[]
5	Colour mode correct; no unintended RGB elements remain	[]
6	Overprint and knockout settings verified in separation preview	[]
7	All placed images at adequate effective resolution at final size	[]
8	All links present and packaged; fonts embedded or outlined as required	[]
9	Minimum type size respected, particularly for statutory text	[]
10	Barcode correct, at valid magnification, with quiet zones intact	[]
11	All mandatory statutory content verified against the approved content list	[]
12	Glue areas free of varnish, lamination and heavy ink coverage	[]
13	File named with correct version and PRODUCTION status token	[]
14	Written approval of this version recorded before release	[]

4.9 Expected Outcomes

This section records deliverables, observations arising from practice, challenges encountered, a proposed evaluation framework and the competencies developed. The status of each category of claim is stated explicitly: observations are qualitative practitioner observations, and no figure in this section is presented as a measured result.

4.9.1 Deliverables

Table 4.14: Summary of deliverables (to be completed)

Work stream	Deliverable type	Quantity	Status at end of internship
WS-1	Dielines constructed or adapted	[]	[]
WS-1	Packaging artwork sets	[]	[]
WS-1	Inserts, stickers, secondary components	[]	[]
WS-2	Physical prototypes built	[]	[]
WS-3	Instruction manuals / assembly guides	[]	[]
WS-4	DFM reviews participated in	[]	[]

Work stream	Deliverable type	Quantity	Status at end of internship
WS-5	Digital creatives and print collateral	[]	[]

TO COMPLETE: Fill the quantities and status, and state plainly what went to production, what was approved but unreleased, and what was shelved. Work that was shelved is not a failure and should be reported — in a commercial studio a substantial proportion of output never reaches production. Avoid claiming commercial outcomes such as sales impact, which you are not in a position to verify.

4.9.2 Observations from practice

1. Physical verification detects a class of error that screen review does not — proportion, fold-crossing artwork and legibility at true size were repeatedly caught on paper mock-ups after passing screen review.
2. The cost of a change is a function of the stage at which it is made: hours at concept, a production cycle after tooling commitment.
3. Constraints consulted early become design parameters; constraints discovered late become revisions.
4. Mandatory content placed last is always placed badly, because it occupies whatever space the creative layout has left over.
5. An instruction guide cannot be validated by its author; only observation of a naive assembler reveals an ambiguous step.
6. In a growing brand a reusable template repays itself across the range, while a one-off file does not.

4.9.3 Challenges encountered

Table 4.15: Challenges encountered and responses adopted

Challenge	Response adopted
Screen-to-print colour shift on brand-critical elements	Working in the destination colour space from the outset; brand colours specified as spot references with documented equivalents
Artwork distorting or breaking across folds	Panel-by-panel development on the dieline; mandatory physical fold before design review
Manufacturing constraints surfacing after artwork completion	Consulting manufacturing at structural definition rather than at review
Instruction steps clear to the author but ambiguous to a first-time assembler	Naive-assembler validation; one action per step; fixed viewpoint
Statutory content competing with creative layout for space	Mandatory content placed on the dieline before creative layout
Risk of an unapproved file reaching a vendor	Status tokens in file names; only PRODUCTION-status files released
Vendor lead times compressing available iteration time	Front-loading low-fidelity iteration; escalating fidelity only when required

TO COMPLETE: Add challenges specific to your experience — software you had to learn, a failed print run, a prototype that would not assemble, a brief that changed mid-project. Specific difficulties honestly reported are the most persuasive evidence that the internship was substantive.

4.9.4 Proposed evaluation framework

Quantitative measurement was not part of the internship. The framework below specifies what would be measured were the design function evaluated systematically. Every metric is marked as proposed and not measured, and none is reported as an observed value.

Table 4.16: Proposed evaluation metrics (not measured during this internship)

Metric	Method of measurement	Status
Revision count to approval	Count from the version-control archive	Proposed
Pre-press defect rate per file released	Vendor pre-flight reports	Proposed
Assembly success rate without assistance	Structured observation with a defined sample	Proposed
Cube utilisation and dimensional weight	Dimensional calculation	Proposed
Template reuse rate across new projects	Asset library audit	Proposed

Revision count to approval and assembly success rate would be the most informative and the least expensive to institute, since both derive from records that already exist or from a low-cost observational protocol.

4.9.5 Competencies developed

Table 4.17: Competencies developed during the internship

Category	Competency
Technical	Dieline construction and adaptation; line-type conventions; bleed, safety zone and glue flap design
Technical	Print production literacy — colour space management, separations, substrate and finish specification, pre-press verification
Technical	Physical model making and additive prototyping
Technical	Design-for-manufacturing analysis against tooling, material and assembly constraints
Domain	Labelling and conformity obligations for toys, and the principal standards relevant to export
Domain	Logistics economics — cube utilisation, dimensional weight, master carton fit
Professional	Working to production deadlines set by vendor lead times; operating within a confidentiality undertaking
Professional	Version control, documentation discipline and written approval records

4.9.6 Relationship to the Computer Science and Engineering programme

The internship was undertaken in a design and manufacturing context while the academic programme is in computer science and engineering. The connections are substantive rather than rhetorical.

- Constraint satisfaction. Packaging design is a constraint satisfaction problem with interacting hard and soft constraints — dimensional, material, cost, regulatory and aesthetic — and the reasoning parallels constrained optimisation.
- Release management. The naming, versioning and release discipline applied to design files is structurally identical to source control practice, with the same failure mode: an unapproved artefact reaching production.
- Usability engineering. Assembly guide design is an information architecture problem, and naive-assembler validation is a usability test conducted on a physical artefact.
- Pipeline and gate design. Establishing a workflow with defined gates, exit criteria and feedback loops is process design, and the same reasoning governs build pipelines and quality gates in software delivery.
- Computational geometry and automation. Dieline construction is parametric and geometric, and is increasingly automated — directly relevant to computational design tooling and to scripted pre-press verification.

TO COMPLETE (recommended): Name the specific subjects from your syllabus that connected to this work — for example Human–Computer Interaction, Software Engineering, Computer Graphics or Design Thinking. Evaluators often ask an interdisciplinary intern to justify the placement, and naming actual courses answers that directly.

4.9.7 Conclusion

The central professional learning from this internship is that in a manufacturing context the quality of a design outcome is governed less by the merit of individual creative decisions than by the order in which constraints are admitted into the process. A workflow that establishes structural, manufacturing and regulatory constraints before creative development, and that verifies decisions physically at low cost before escalating fidelity, produces fewer revisions and fewer late failures than one treating those constraints as review criteria applied to a finished proposal. This differs materially from the way design problems are framed in coursework, where the constraint set is usually given rather than discovered and where a late change costs a revision rather than a production run.

Future scope for the organisation includes a parametric dieline library reducing structural definition time on new projects, a documented brand and packaging design system, a pictogram library giving the range a consistent instructional language,

institution of the metrics proposed in Section 4.9.4, sustainable packaging evaluation through mono-material construction and reduced void volume, and computational tooling for repetitive tasks such as automated dieline generation from product dimensions and scripted pre-press verification.

4.10 Certificates of Completion and Communication Proof

The following documents are required by the prescribed report format and are to be attached at this point.

Table 4.18: Documents to be attached

#	Document	Attached
1	Internship completion certificate issued by Intelligentia Labs Pvt. Ltd. on company letterhead, signed	[]
2	Copy of the appointment letter dated for the position of Graphics and Prototyping Intern	[]
3	Email correspondence confirming commencement and completion of the internship	[]
4	Weekly logbook or attendance record countersigned by the reporting manager, if required	[]
5	Written permission from the organisation for any proprietary images reproduced in this report	[]

ACTION REQUIRED — COMPLETION CERTIFICATE: This must be issued and signed by the company; it cannot be prepared by the student. Request it from your reporting manager or from the Director, InventoJoy, giving your name, role and exact start and end dates. Insert the scanned certificate on the dedicated unnumbered page earlier in this report and again here if your department asks for it in both places.

ACTION REQUIRED — COMMUNICATION PROOF: Attach screenshots or printouts of the offer email and the completion correspondence. Redact any stipend figures if your department prefers financial terms omitted, and redact any third-party personal contact details.

5. BIBLIOGRAPHY / REFERENCES

The structure below is provided for completion. No citation is presented as verified. Each entry must be completed with the actual title, publisher, edition or version and access date of the source consulted, and any entry not actually consulted must be deleted before submission.

ACTION REQUIRED: Replace each bracketed entry with a source you have actually read, formatted in the citation style your department requires. Delete the rest. A short accurate reference list is always better than a long invented one.

Standards and regulatory sources

- [1] Bureau of Indian Standards. Indian Standard on safety of toys — insert exact designation, part, edition year and title as verified from the BIS catalogue.
- [2] Government of India, Department for Promotion of Industry and Internal Trade. Toys (Quality Control) Order — insert exact title, year and gazette reference as verified.
- [3] Destination-market toy safety standards relevant to export work — insert exact designations and years as verified.
- [4] Packaged commodity labelling rules applicable to retail packaging — insert exact title and year as verified.

Technical references

- [5] Reference work on packaging structural design and carton construction — author, title, edition, publisher, year.
- [6] Reference work on print production and pre-press for designers — author, title, edition, publisher, year.
- [7] Reference work on design for manufacturing and assembly — author, title, edition, publisher, year.

Software documentation and organisational sources

- [8] Documentation for each design and CAD application used — publisher, version, URL, date accessed.
- [9] Intelligentia Labs Pvt. Ltd. Appointment letter for the position of Graphics and Prototyping Intern, InventoJoy. Summarised in Table 4.1.
- [10] InventoJoy brand guidelines and vendor specification sheets — internal documents; cite only if permitted by the organisation.